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BASIC MATHEMATICS

FORM TWO

Form Two National Assessment (FTNA) 2025 Answer Key and Worked Solutions

Subject	Basic Mathematics
Code	041
Level	Form Two
Year	2025
Exam Board	NECTA
Exam	Form Two National Assessment (FTNA)
Type	Answer key and worked solutions
Questions	10 compulsory questions; 24 answered parts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: BasicMath-F2-2025.pdf

Question 2(a) omits the overlap between tea and coffee drinkers. Both the intended disjoint-group answer and the general expression are explained. Question 8(a) requires a corrected vertex order when writing the similarity statement.

Question 1 — Arithmetic and rounding

1(a). Evaluate $(1/5)$ of $[(50 \div 5 + 5) - (8 \times 4 - 2)]$.

Step 1: Inside each bracket perform division or multiplication first: $50 \div 5 + 5 = 10 + 5 = 15$; $8 \times 4 - 2 = 32 - 2 = 30$.

Step 2: The bracket difference is $15 - 30 = -15$. Multiply by $1/5$: $-15/5 = -3$.

Final Answer: -3 .

1(b). Town populations are 65,600, 13,400 and 29,700, given to the nearest hundred. Approximate their total to the nearest thousand.

Step 1: Add the stated populations first: $65,600 + 13,400 + 29,700 = 108,700$.

Step 2: The hundreds digit is 7, so round 108,700 up to 109,000.

Final Answer: 109,000 people.

Section Summary: Apply the stated mathematical rule and show intermediate calculations. In geometry, match corresponding vertices carefully; in sets, account for any overlap.

Question 2 — Fractions and percentages

2(a). Of 15 children, $2/5$ drink tea and $1/3$ drink coffee. How many drink neither?

Step 1: Tea drinkers = $(2/5) \times 15 = 6$; coffee drinkers = $(1/3) \times 15 = 5$.

Step 2: If the groups are disjoint, drinkers = $6 + 5 = 11$, leaving $15 - 11 = 4$.

Step 3: In general, tea or coffee = $6 + 5 - t$, so neither = $15 - (11 - t) = 4 + t$.

Final Answer: 4 children if no child drinks both; with an overlap of t children, the answer is $4 + t$.

Study Tip: The question does not state whether the groups overlap. The usual intended arithmetic answer assumes no overlap; otherwise an exact count cannot be determined from the given facts.

2(b). Of 40 students, 25 passed English. Find the percentage who failed.

Step 1: Failed = $40 - 25 = 15$.

Step 2: Percentage failed = $(15/40) \times 100 = 37.5\%$.

Final Answer: 37.5%.

Section Summary: Apply the stated mathematical rule and show intermediate calculations. In geometry, match corresponding vertices carefully; in sets, account for any overlap.

Question 3 — Profit, loss and ratio

3(a). A textbook bought for Tsh 8,500 was sold for Tsh 6,000. Find the percentage loss.

Step 1: Loss = $8,500 - 6,000 = 2,500$.

Step 2: Loss percentage uses cost price: $(2,500/8,500) \times 100 = 500/17\% \approx 29.4118\%$.

Final Answer: $29 \frac{7}{17}\% \approx 29.41\%$.

3(b)(i). Share 33 kg of meat among A,B,C,D in ratio 4:5:6:7. Find the largest share.

Step 1: Total ratio parts = $4 + 5 + 6 + 7 = 22$.

Step 2: One part = $33/22 = 1.5$ kg. Largest share = $7 \times 1.5 = 10.5$ kg.

Final Answer: 10.5 kg.

3(b)(ii). Which family receives the largest share?

Step 1: Compare the ratio parts 4,5,6,7.

Step 2: The largest part, 7, belongs to D.

Final Answer: Family D.

Section Summary: Apply the stated mathematical rule and show intermediate calculations. In geometry, match corresponding vertices carefully; in sets, account for any overlap.

Question 4 — Angles and perimeter

4(a). $AB \parallel PQ$ and RS is a transversal. Find a, b, w, x, y, z using the given 60° and 120° .

Step 1: At PQ , a is vertically opposite the given 60° , so $a = 60^\circ$. The adjacent angles b and w each equal $180 - 60 = 120^\circ$.

Step 2: At AB , y is vertically opposite the given 120° , so $y = 120^\circ$. Each adjacent angle x and z is $180 - 120 = 60^\circ$.

Step 3: Corresponding angles across the parallel lines confirm the same acute and obtuse values.

Final Answer: $a = 60^\circ$, $b = 120^\circ$, $w = 120^\circ$, $x = 60^\circ$, $y = 120^\circ$, $z = 60^\circ$.

4(b). Triangle ABC has perimeter 16 cm, $AB = (5 + x)$ cm, $AC = (2 + x)$ cm and $BC = 5$ cm. Find x , AB and AC .

Step 1: Add the sides: $(5 + x) + (2 + x) + 5 = 16$.

Step 2: $12 + 2x = 16$, so $x = 2$. Substitute into each variable side.

Final Answer: $x = 2$; $AB = 7$ cm; $AC = 4$ cm.

Section Summary: Apply the stated mathematical rule and show intermediate calculations. In geometry, match corresponding vertices carefully; in sets, account for any overlap.

Question 5 — Algebra

5(a)(i). Simplify $7m - 2(5n - 4m) + 11n - 5m$.

Step 1: Expand the bracket with its negative multiplier: $-2(5n - 4m) = -10n + 8m$.

Step 2: Collect m : $7m + 8m - 5m = 10m$. Collect n : $-10n + 11n = n$.

Final Answer: $10m + n$.

5(a)(ii). Simplify $(5x - 2y) - 2(4x - 3y)$.

Step 1: Expand: $5x - 2y - 8x + 6y$.

Step 2: Collect like terms: $5x - 8x = -3x$ and $-2y + 6y = 4y$.

Final Answer: $-3x + 4y$.

5(b). Two numbers sum to 19 and differ by 5. Find them.

Step 1: Let the larger number be a and smaller b : $a + b = 19$, $a - b = 5$.

Step 2: Add equations: $2a = 24$, so $a = 12$. Then $b = 19 - 12 = 7$.

Final Answer: 12 and 7.

Section Summary: Apply the stated mathematical rule and show intermediate calculations. In geometry, match corresponding vertices carefully; in sets, account for any overlap.

Question 6 — Straight lines

6(a)(i). The line $ky = kx + x + 7$ has gradient 2. Find k .

Step 1: For $k \neq 0$, divide by k : $y = [(k + 1)/k]x + 7/k$.

Step 2: Set gradient $(k + 1)/k = 2$. Then $k + 1 = 2k$, so $k = 1$.

Final Answer: $k = 1$.

6(a)(ii). Write the y-intercept of the line.

Step 1: Substitute $k = 1$: $y = 2x + 7$.

Step 2: At $x = 0$ the y-coordinate is 7.

Final Answer: 7, at the point (0,7).

6(b). $Kx - My - 4 = 0$ passes through $A(-2,5)$ and $B(2,-3)$. Find K and M .

Step 1: Substitute A: $-2K - 5M = 4$. Substitute B: $2K + 3M = 4$.

Step 2: Add the equations: $-2M = 8$, giving $M = -4$.

Step 3: Then $2K + 3(-4) = 4$ gives $2K = 16$ and $K = 8$. Check both points in $8x + 4y - 4 = 0$.

Final Answer: $K = 8$; $M = -4$.

Section Summary: Apply the stated mathematical rule and show intermediate calculations. In geometry, match corresponding vertices carefully; in sets, account for any overlap.

Question 7 — Indices and radicals

7(a). Evaluate $8^{(-2/3)} + (3^{-2})/(2^{-3})$.

Step 1: $8^{(-2/3)} = 1/(\sqrt[3]{8})^2 = 1/4$. Also $3^{-2} = 1/9$ and $2^{-3} = 1/8$.

Step 2: The quotient is $(1/9)/(1/8) = 8/9$.

Step 3: Add with denominator 36: $1/4 + 8/9 = 9/36 + 32/36 = 41/36$.

Final Answer: $41/36 = 1 \frac{5}{36}$.

7(b). Rationalise $\sqrt{2}/(\sqrt{3} + \sqrt{2})$.

Step 1: Multiply numerator and denominator by the conjugate $\sqrt{3} - \sqrt{2}$.

Step 2: Numerator = $\sqrt{6} - 2$; denominator = $(\sqrt{3})^2 - (\sqrt{2})^2 = 3 - 2 = 1$.

Final Answer: $\sqrt{6} - 2$.

Section Summary: Apply the stated mathematical rule and show intermediate calculations. In geometry, match corresponding vertices carefully; in sets, account for any overlap.

Question 8 — Similarity and congruence

8(a). In right triangle ABC, $A = 90^\circ$, $AD \perp BC$ and D lies on BC. Prove the large and smaller triangle ABD are similar.

Step 1: $\angle BAC = \angle BDA = 90^\circ$.

Step 2: $\angle ABC = \angle DBA$ since BD lies on BC. Two corresponding angles match, establishing similarity.

Final Answer: $\triangle ABC \sim \triangle DBA$ by AA.

Study Tip: The paper names the two triangles as ABC and ABD without matching the order. The correct ordered correspondence is $A \leftrightarrow D$, $B \leftrightarrow B$, $C \leftrightarrow A$.

8(b). BE bisects $\angle DEF$ and $DE = FE$. Prove $\triangle BEF \cong \triangle BED$.

Step 1: $EF = ED$ is given and BE is a common side.

Step 2: Because BE bisects the angle at E , $\angle BEF = \angle BED$. The equal angle lies between the two equal sides, so SAS applies.

Final Answer: $\triangle BEF \cong \triangle BED$ by SAS.

Section Summary: Apply the stated mathematical rule and show intermediate calculations. In geometry, match corresponding vertices carefully; in sets, account for any overlap.

Question 9 — Trigonometry and Pythagoras

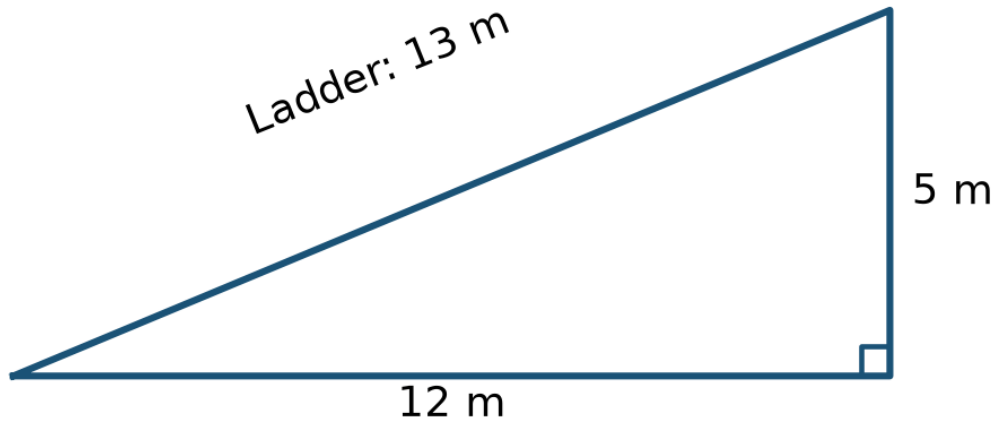
9(a). In the right triangle, the hypotenuse is 30 cm and the opposite side to 70° is y . Find y to one decimal place.

Step 1: $\sin 70^\circ = \text{opposite/hypotenuse} = y/30$.

Step 2: $y = 30 \sin 70^\circ \approx 30 \times 0.9396926 = 28.1908 \text{ cm}$; round to 28.2 cm.

Final Answer: 28.2 cm.

9(b)(i). Draw a ladder leaning against a 5 m vertical wall, with its foot 12 m from the wall.



Step 1: Draw the wall perpendicular to the ground and label their meeting point as a right angle.

Step 2: Join the wall's top to the foot of the ladder; label the vertical and horizontal distances.

Final Answer: A right triangle with vertical side 5 m, horizontal side 12 m and the ladder as the hypotenuse.

9(b)(ii). Find the ladder length.

Step 1: Pythagoras gives $\text{ladder}^2 = 5^2 + 12^2 = 25 + 144 = 169$.

Step 2: The positive square root of 169 is 13, so the ladder is 13 m long.

Final Answer: 13 m.

Section Summary: Apply the stated mathematical rule and show intermediate calculations. In geometry, match corresponding vertices carefully; in sets, account for any overlap.

Question 10 — Sets and grouped data

10(a). Of 120 students, 40 learn English, 60 Kiswahili, and 30 both. How many learn neither?

Step 1: Union = $40 + 60 - 30 = 70$; subtract the overlap once.

Step 2: Neither = $120 - 70 = 50$.

Final Answer: 50 students.

10(b). Group the 30 History marks into 30-39, 40-49, 50-59, etc.; find the percentage scoring 50-69.

Step 1: Tally each listed mark into exactly one class: counts 7,10,8,2,3. Check their sum is 30.

Step 2: Scores from 50 through 69 lie in two classes: $8 + 2 = 10$ students.

Step 3: Percentage = $(10/30) \times 100\% = 33\frac{1}{3}\%$.

Final Answer: Frequencies: 30-39: 7; 40-49: 10; 50-59: 8; 60-69: 2; 70-79: 3. Total 30. Percentage scoring 50-69 = $33\frac{1}{3}\%$.

Section Summary: Apply the stated mathematical rule and show intermediate calculations. In geometry, match corresponding vertices carefully; in sets, account for any overlap.